



Original software publication

‘Forensic-DataFusion-Tool’: A Python-based application for exploratory forensic data analysis using merged datasets from analytical sensors

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ABSTRACT

Portable sensors for on-site forensic analysis have advanced significantly, enabling reliable methods for crime scene investigation. Non-destructive analytical instruments are especially useful for providing chemical information from the same specimen. Combining data from these instruments through data fusion enhances analytical responses. Data fusion merges data from different sources to improve exploratory and predictive models. No current application supports multi-dataset fusion on a single platform. To address this, we developed a Python-based ‘Forensic-DataFusion-Tool’ to merge raw and preprocessed data from multiple sensors, speeding up data fusion and enabling future machine learning updates, including classification algorithms.

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Software code languages, tools, and services used

Compilation requirements, operating environments & dependencies

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V1.0

<https://github.com/SoftwareImpacts/SIMPAC-2025-118>

<https://codeocean.com/capsule/5127057/tree/v1>

GNU General Public License (GPL)

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Python, Visual Studio Code

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<https://github.com/foselab/Forensic-DataFusion-Tool/blob/main/code/resources/Instruction.pdf>

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1. Motivation and significance

Portable sensors and instruments designed to facilitate on-site analysis of forensic samples have rapidly proliferated in recent years. Moreover, significant attention has been devoted to developing robust, validated methods that can provide reliable analyses on the site at crime scenes [1].

Various analytical instruments, especially non-destructive ones, can provide diverse chemical information about the same sample. However,

appropriately combining the signals from these instruments can create a synergic effect, improving the overall analytical response. This is the principle behind the data fusion (DF) approach. DF is designed to create enhanced exploratory or predictive models, including quantitative and classification models, by merging data from various sources [2].

The data fusion approach can be divided into three different classes, based on the level at which the data are merged [3]:

- Low-level data fusion (LLDF), which merges the raw data from each instrument;

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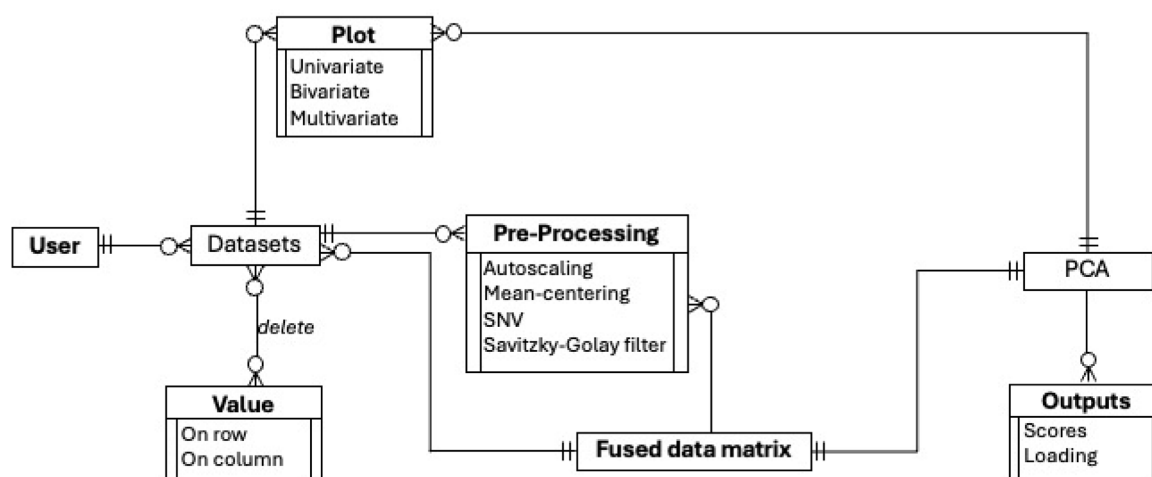


Fig. 1. Workflow in the Forensic-DataFusion-Tool.

- Mid-level data fusion (MLDF), which merges the features extracted from each dataset;
- High-level data fusion (HLDF), which merges the responses of all instruments.

Although various applications for chemical data analysis are available, none currently support LLDF across multiple analytical datasets within a single platform. To address this gap, we have developed a Python-based application called the 'Forensic-DataFusion-Tool'. This tool facilitates comprehensive data exploration through principal component analysis (PCA), enabling the fusion of both raw and preprocessed analytical data.

The Forensic-DataFusion-Tool provides a versatile analysis platform by supporting univariate, bivariate, and multivariate approaches to using data from individual sensors or instruments. It also incorporates various preprocessing methodologies to address variations in feature properties within datasets, such as discrepancies in metrics, absolute values, and variable ranges, which can affect multivariate statistical methods. By standardizing such differences, these preprocessing steps enable more accurate and reliable results [4].

Finally, the Forensic-DataFusion-Tool allows raw and preprocessed data from multiple instruments or sensors to be merged prior to data exploration through PCA. This capability provides a robust platform for easily fusing and exploring the data in the resulting data matrix. As the application is further developed, new machine learning functionalities will be implemented in future versions, such as classification algorithms.

2. Software description

Forensic-DataFusion-Tool was written in Python code (v 2.12.2), within the Visual Studio Code environment. The application has been tested on an HP Desktop M01-F3030 ns PC equipped with an AMD Ryzen 5 5600G processor, 8 GB of RAM, 512 GB of SSD, and an AMD Radeon graphics card. The Python modules were used to create a user-friendly desktop application that runs on Windows computers. The tool features a modular architecture, allowing for easy integration of additional functionalities, such as real-time data processing as well as machine learning and deep learning algorithms. The Forensic-DataFusion-Tool supports the Excel data formats that are typically used to collect analytical data before the data analysis process. The workflow in the application is illustrated in Fig. 1.

The user can upload multiple analytical datasets into the application and examine them in the initial phase of data analysis using a variety of approaches, including univariate, bivariate, and multivariate

methodologies, and their graphical representations as plots. This allows for a comprehensive exploration of a dataset, with univariate analysis focusing on individual variables, bivariate analysis examining the relationships between two variables, and multivariate analysis exploring the interactions and patterns across multiple variables simultaneously. Each of these methods helps users to uncover insights in data at different levels of complexity. Moreover, rows or columns can be deleted from each dataset, depending on the analysis carried out and the user's aims.

Each dataset can be preprocessed with different methodologies, such as autoscaling, means centering and standard normal variate (SNV). More preprocessing methods can be applied to the same datasets one after another to extract as much useful information as possible. The raw or preprocessed data can then be fused into a single data matrix, which can be further explored using PCA. The main outputs of the first version of the application include a matrix of the scores and loadings, along with the corresponding plots. These outputs not only provide valuable insights for data interpretation but can also be used as inputs for further analyses, including the application of machine learning and deep learning methodologies.

3. Preprocessing

Four different preprocessing methodologies have been implemented in the application Forensic-DataFusion-Tool: autoscaling, mean centering, SNV, and Savitzky-Golay smoothing (see Table 1).

Mean centering can be applied to columns in a dataset by subtracting the mean of each column (or variable) from its corresponding values. This transform the data so that each column has a mean of zero [5].

Autoscaling eliminates the influence of different units of the variables by subtracting the mean of each variable and dividing the result of this subtraction by the standard deviation. Autoscaled variables all have unit variance (unit variance scaling) and are equally important for model building [6].

SNV preprocessing is typically used for spectroscopic data and involves centering and scaling each spectrum based on its standard deviation to mitigate scattering effects [7]. Fig. 2 shows an example of a plot obtained with the application, comparing infrared (IR) spectra before and after the preprocessing.

The Savitzky-Golay method is commonly used for smoothing/noise reduction based on local least-squares polynomial approximation. Specifically, when a Savitzky-Golay filter is applied to a signal, it generates a polynomial P with coefficients corresponding to a degree of

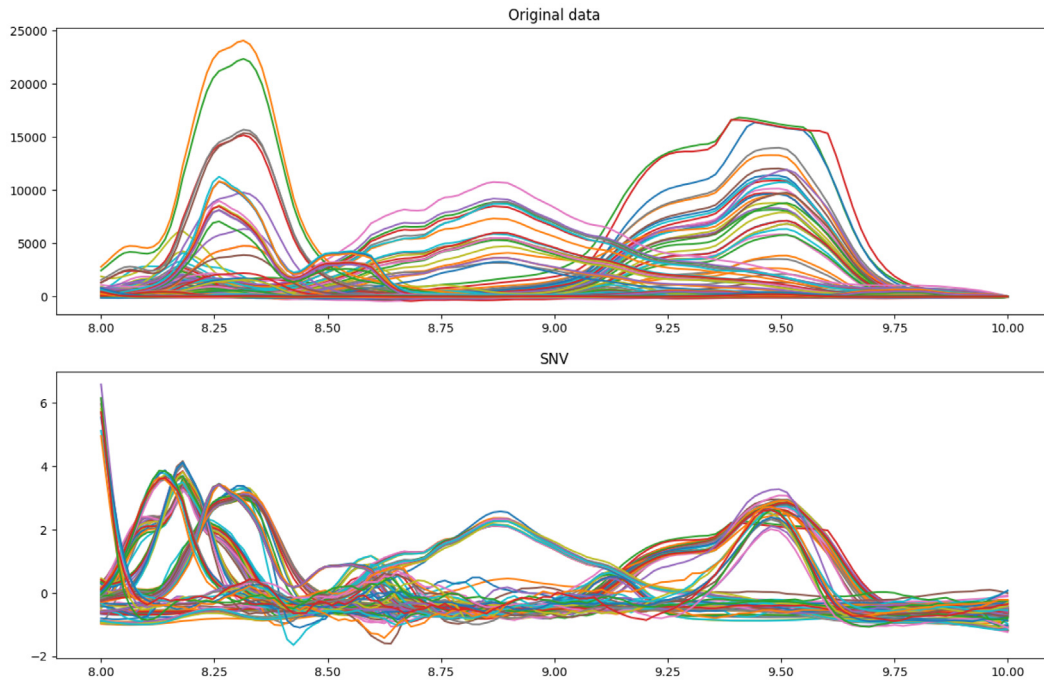


Fig. 2. A plot of preprocessed infrared spectra.

Table 1
Formula of pre-processing methods implemented in the application.

Method	Formula
Mean centering	$\tilde{x}_{ij} = x_{ij} - \bar{x}_i$
Autoscaling	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{s_j}$
Standard normal variate	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{(x_{imax} - x_{imin})}$
Definitions	
$\bar{x}_i$	Mean of the column
$\bar{x}_j$	Mean of the row
s_i	Standard deviation: $s_i = \sqrt{\frac{\sum_{j=1}^J (x_{ij} - \bar{x}_i)^2}{J-1}}$

k (the degree of the polynomial). The polynomial is fitted to N points of the signals as follow:

$$N = N_r + N_l + 1$$

In this equation N is the window size, and N_r and N_l are the numbers of points from the center of the signal, respectively, on the right and left of it.

The obtained coefficients from this fitting procedure are used to estimate the new value at the specific wavelength or variable considered. The degree of the derivative has a direct impact on the necessary number of points in the smoothing window; higher degrees require more points. In any case, the second derivative eliminates the baseline effect and the offset [8–10].

4. Principal component analysis

PCA is an unsupervised pattern recognition technique used to decrease the dimensions of the original dataset and reduce the noise while keeping most of the useful information. Following the application of the method, a new set of uncorrelated variables is obtained, which are known as the Principal Components (PCs). These are linear combinations in the original variables; the first PC accounts for the largest variance of the dataset, the second one accounts for the largest residue

variance, and so on. The values that characterize each sample in the PC space defined by the PCs are called scores.

They can be used as input for other multivariate techniques [11,12].

PCA mathematically decomposes the original data matrix X into the sum of the outer product of vectors s (scores) and l (loadings) plus a residual matrix E . [13] as follow:

$$X = s_1 \times l_1 + s_2 \times l_2 + \dots s_k \times l_k + E$$

Fig. 3 shows the graphical user interface (GUI) used to perform the PCA in the application.

Unsupervised pattern recognition also allows the outliers to be detected within the dataset. For this purpose, Hotelling T^2 and Q residuals were implemented in the application. Hotelling T^2 is a generalization of Student's t that is used in multivariate hypothesis testing [14]. Data points that are closer to the center of the class distribution have lower Hotelling T^2 values and vice versa [15]. The Q residual indicates how well each sample conforms to the PCA model. It is a measure of the difference or residual between a sample and its projection into the principal components [16].

Samples with low Q-residuals are well represented by the model, i.e. they have small (orthogonal) distances between their low-dimensional projections [17].

5. Impact and future development

The Forensic-DataFusion-Tool application has played an important role in analyzing data for the European project RISEN (Real-time on-site forensic trace qualification; HORIZON2020, Grant Agreement No 883116) [18]. It was effectively used to examine analytical data from two sensor prototypes: a gas chromatograph coupled to a quartz-enhanced photoacoustic spectroscopy (GC-QEPAS) [19], and an ion mobility spectrometer (IMS) [20]. Additionally, this tool has been used to explore data from individual sensors, such as GCQEPAS, IMS, and IR, both independently and through low-level DF [21]. Initially, the application helped identify the best preprocessing techniques for each type of data. Subsequently, data fusion was employed to extract more valuable insights. For example, after the DF, the PCA plots revealed more compact and well-separated clusters in hyperspace for the classes

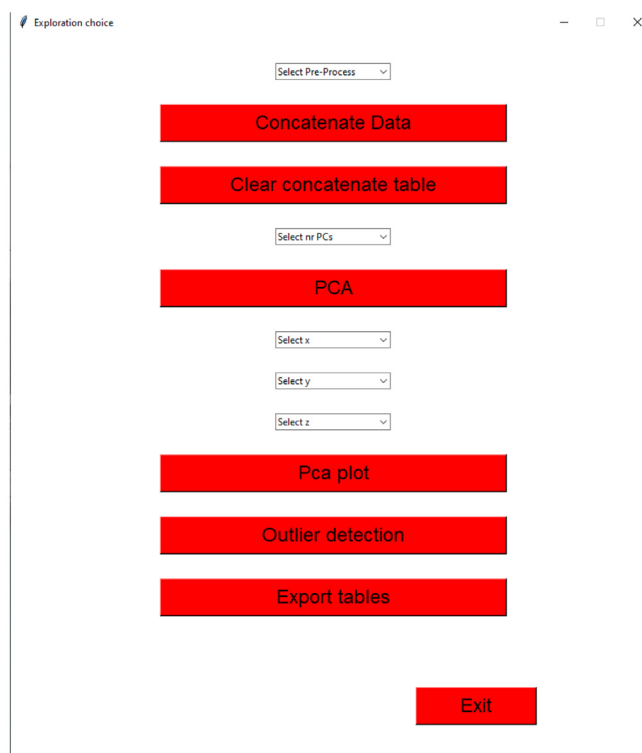


Fig. 3. GUI for PCA.

under investigation, including explosives and their precursors, chemical warfare agent simulants, drugs, and their precursors. Moreover, compared to traditional machine learning applications, in which data cannot be directly fused within the application, this process significantly reduced the computational time. The application has also been used to explore data from GC-QEPAS sensors prior to applying deep learning algorithms [22].

Being entirely developed in Python, the application benefits from the computational efficiency and versatility of established scientific libraries (e.g., NumPy, pandas, scikit-learn, matplotlib), while providing a graphical, user-friendly interface that eliminates the need for scripting. Users can import datasets, apply multiple preprocessing steps, perform low-level data fusion, and generate PCA plots directly from the interface, making complex data analysis workflows accessible to non-programmers. In contrast, most existing commercial software packages supporting chemometric analysis are license-based and expensive, while open-source alternatives such as R or Python scripts require advanced programming knowledge. The Forensic-DataFusion-Tool fills this gap by offering a free, intuitive, and easily deployable desktop solution.

Preliminary benchmarking performed during the RISEN project indicates that the integrated data-fusion workflow can reduce computational time by 70%–80% compared with equivalent analyses carried out via traditional Python scripting, and by 30%–40% compared to standalone machine learning tools in the case of data fusion, while maintaining comparable clustering quality and reproducibility. Although these results are illustrative, they demonstrate the potential of the tool to accelerate data analysis in forensic workflows.

The modular structure of the application also enables on-site deployment. When installed directly on the computer or embedded system associated with the analytical sensor, the Forensic-DataFusion-Tool can perform data preprocessing, fusion, and PCA in real time at the crime scene, without the need for external computation or network connection. This capability significantly enhances the operational efficiency

of portable forensic laboratories, supporting real-time decision-making and preliminary classification of unknown substances directly in the field.

In terms of adoption potential, the application is aimed at forensic laboratories, analytical chemistry research groups, and field-deployable units that require rapid data integration and visualization. Its modular design facilitates the inclusion of additional functionalities, such as real-time data processing, machine learning classification, and deep learning algorithms, which are planned for future releases.

Some limitations should be noted. While the current Python-based implementation of the application offers high flexibility, ease of integration, and strong performance for small to medium-sized datasets, it is not yet optimized for very large-scale data processing. Handling datasets larger than 10^4 spectra can lead to significant increases in memory usage and processing time, due to Python's inherent memory management and single-threaded performance limitations. Additionally, classification modules are not yet available, and regression functionalities are still under development. Future updates will focus on improving scalability through optimized data handling, enabling on-device processing for edge applications, implementing automated model validation, and expanding visualization tools to improve both robustness and user accessibility.

The current version of the application supports data analysis up to the data exploration phase but does not incorporate the classification models typically used in this field. More functionalities, such as machine learning classification methods and regression models, will be implemented in subsequent versions. The authors believe that this application fills a gap in the current commercial tools, which do not offer capabilities for DF within the platform.

6. Conclusion

The Forensic-DataFusion-Tool application was developed and tested during the European RISEN project. Its effective use in analyzing data from diverse sensor prototypes, including GC-QEPAS, IMS, and IR, demonstrates its utility in integrating and exploring complex datasets through PCA.

The tool's current capabilities, which encompass data exploration but exclude classification models, fill a notable gap in the existing commercial applications, which lack integrated DF functionality. Thus, it represents a significant advancement in data analysis. By providing a robust platform for merging and analyzing data from multiple sources, the Forensic-DataFusionTool offers a valuable solution for forensic and analytical applications. Future developments to enhance its functionality, beginning with advanced machine learning techniques.

CRedit authorship contribution statement

Giorgio Felizzato: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Michele Verdi:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Angelo Michele Gargantini:** Writing – review & editing, Software, Resources, Methodology, Conceptualization. **Nico Pellegrinelli:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Francesco Saverio Romolo:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Francesco Saverio Romolo reports financial support was provided by Horizon Europe. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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